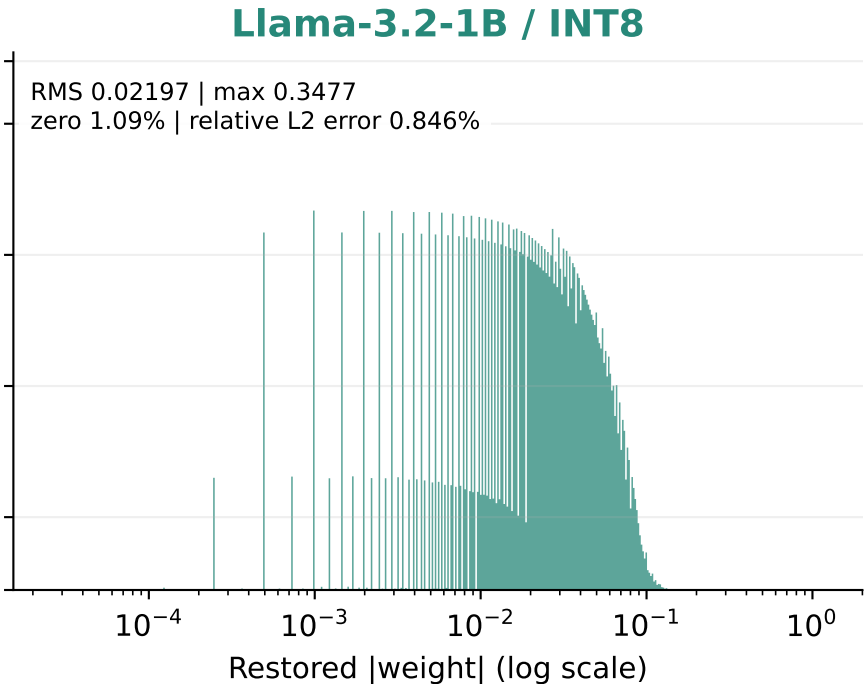
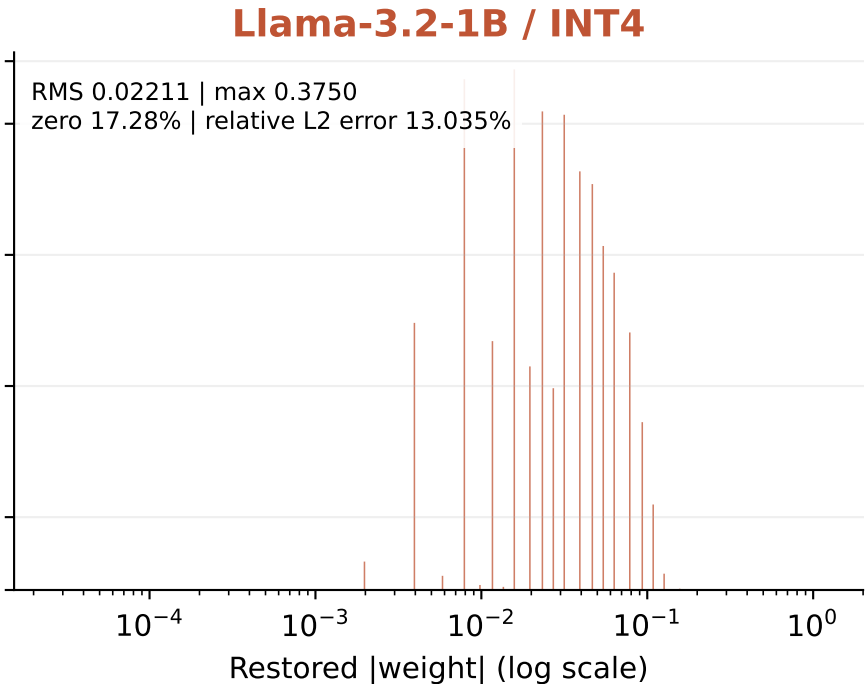
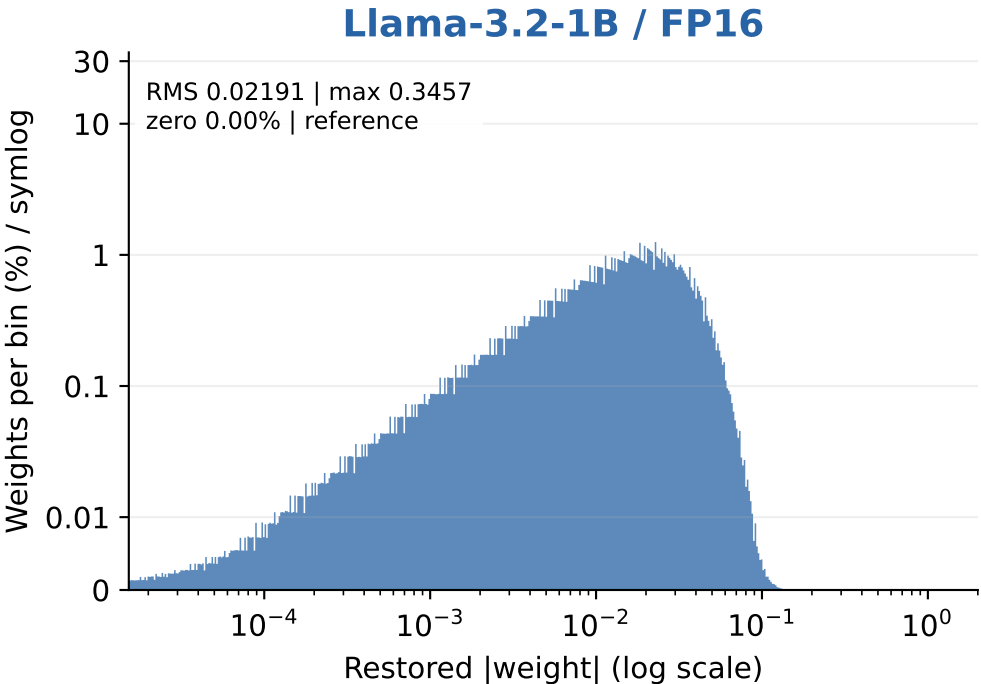
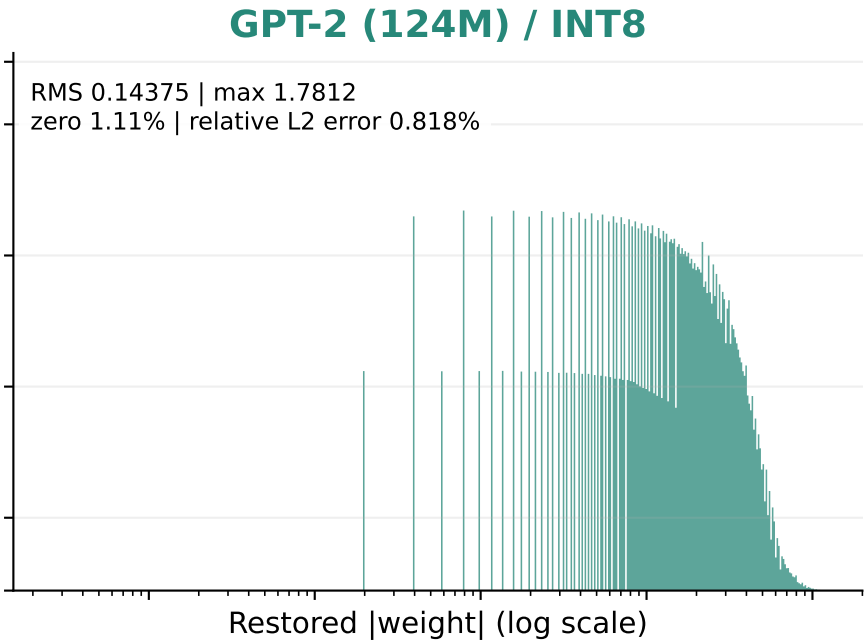
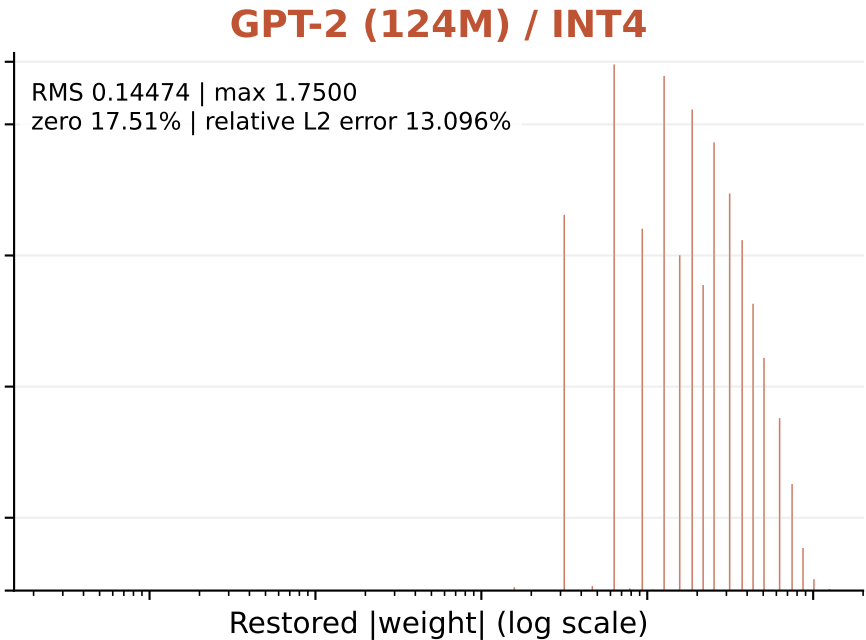
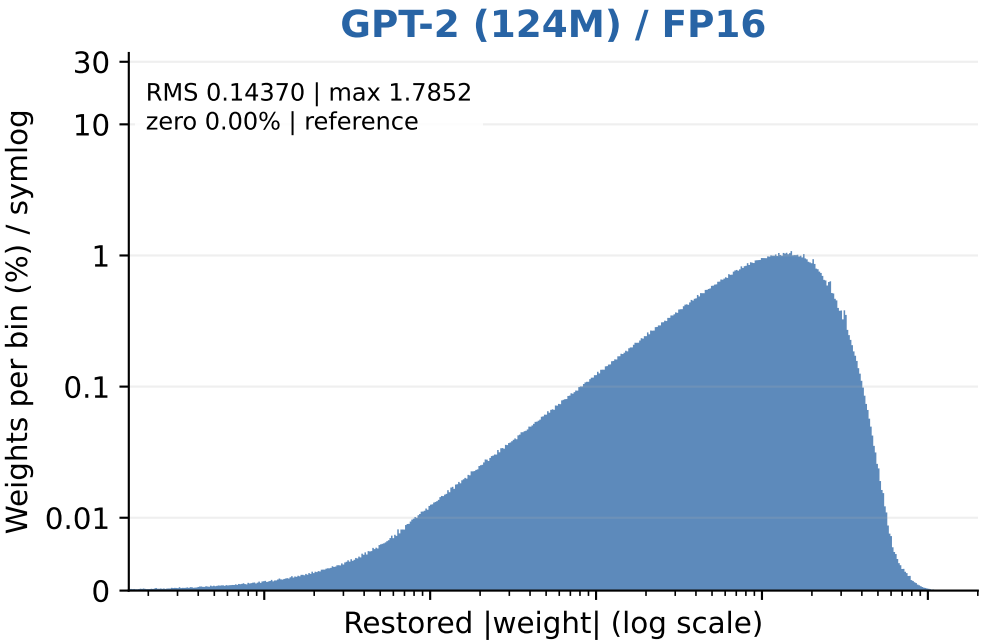


LM head weights: FP16 vs current PoTal HP1

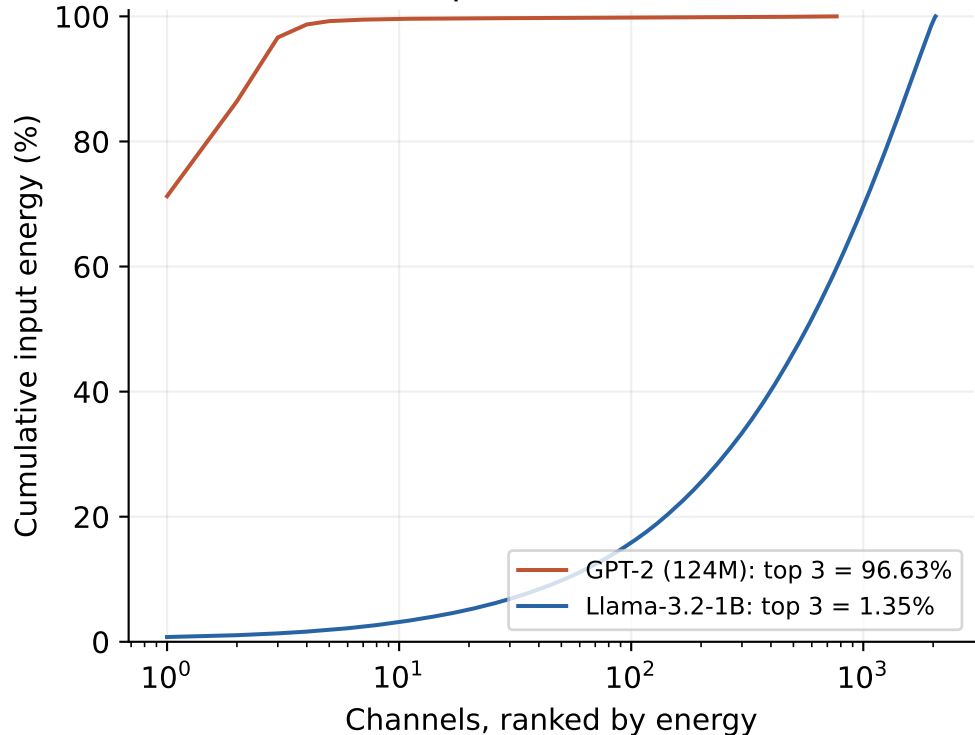
Shared token embedding / output matrix. All weight values; 1/32-log2 magnitude bins; no smoothing.



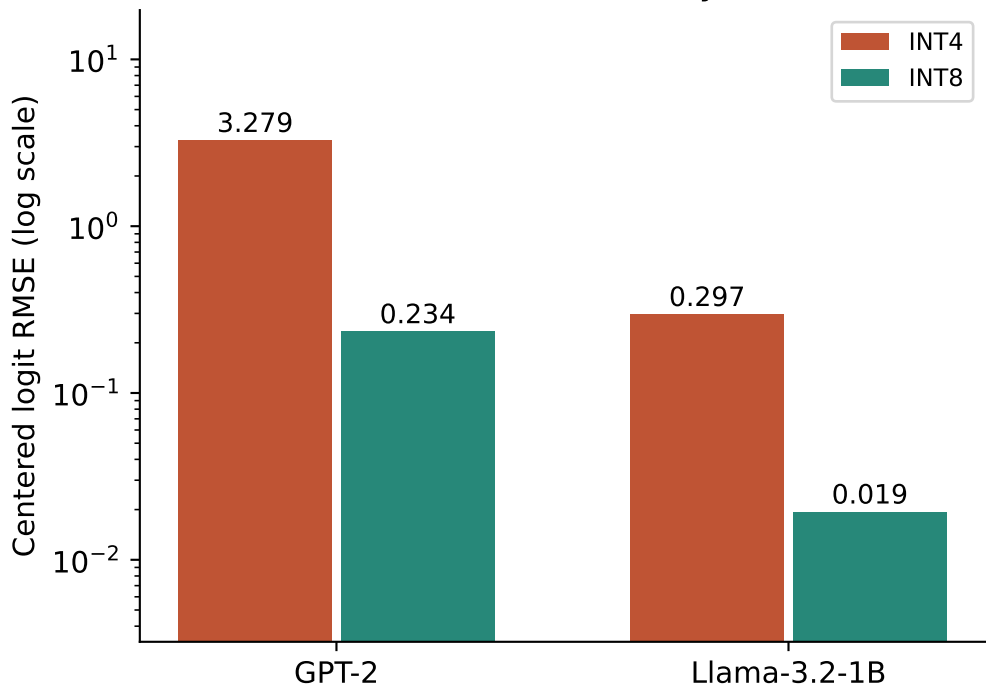
Zeros are reported separately. Shared axes across all six panels. Full histograms are retained in data/*-magnitude-fine.csv.
Similar relative weight error does not imply similar logit error: the input activations and their direction also matter.

Why GPT-2's LM head is sensitive: three separate measurements

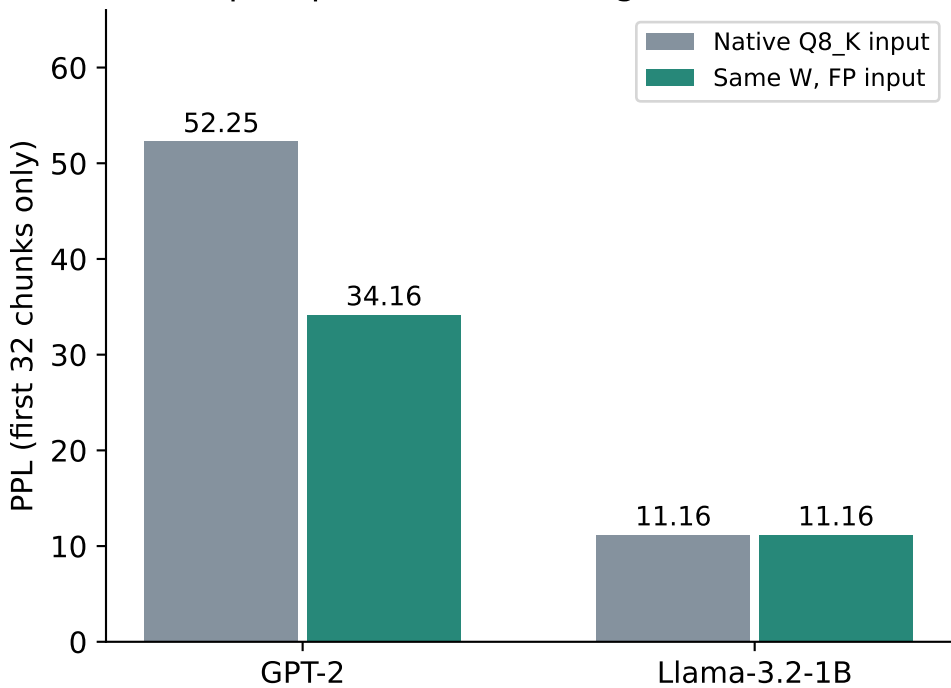
Historical CPU head inputs
8,192 captured rows / model



Current HP1 weights, fixed historical H
128 rows, entire vocabulary, FP64 dot



Historical CPU control: same Q6_K W
Input quantization changes; H is fixed



H was captured from Q4_0-body / Q6_K-head CPU runs in September 2026. It is not a current PoTal activation trace.
Middle: weight-only sensitivity replay; no ExSIA, RMD, SCU or PPL evaluation. Right: 8,160 scored tokens, not full-dataset PPL.
GPT-2 control: restoring native Q8_K-rounded H in the FP head restores PPL 52.2543. Current PoTal causality remains untested.